```
#Logistic Regression (Classification)

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
# data loading
dataset=pd.read_csv("Placement selection dataset.csv")
dataset
```

Next steps: Generate code with `dataset` New interactive sheet

	Age	Gender	Stream	Internships	CGPA	Hostel	HistoryOfBacklogs
0	22	Male	Electronics And Communication	1	8	1	1
1	21	Female	Computer Science	0	7	1	1
2	22	Female	Information Technology	1	6	0	0
3	21	Male	Information Technology	0	8	0	1
4	22	Male	Mechanical	0	8	1	0
5	22	Male	Electronics And Communication	0	6	0	0
6	21	Male	Computer Science	0	7	0	1
7	21	Male	Information Technology	1	7	0	0
8	21	Male	Computer Science	2	6	0	0
9	21	Female	Computer Science	1	6	1	0
10	22	Male	Computer Science	1	7	0	0
11	22	Female	Electrical	1	8	0	1
12	21	Female	Computer Science	2	6	1	1
13	21	Male	Computer	1	8	0	1

```
#one-hot encoding
data=pd.get_dummies(dataset,drop_first=True)
```

15	22	Male	Mechanical	0	8	1	0

```
#data split input and output
independent=data[['Age', 'Internships', 'CGPA', 'Hostel', 'HistoryOfBacklogs
dependent=data[['PlacedOrNot']]
```

17	21	Male	Civil	0	6	0	0

```
#training and testing
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(independent,dependent,test_si
```

19	22	Male	Science	1	7	0	0

```
#Model creation
from sklearn.linear_model import LogisticRegression

classifier = LogisticRegression()
classifier.fit(x_train, y_train.values)
```

```
weight=classifier.coef_
print("Weights (Coefficients):", weight)
bias=classifier.intercept_
print("Bias (Intercept):", bias)
y_pred=classifier.predict(x_test)
y_pred
```

```
Weights (Coefficients):
 [[ 0.22493713  0.96691616  1.52163818  0.30547619  0.14087174  0.01351093
  -0.80362593  0.29595243  0.15156951  0.73140602  0.37692861]]
Bias (Intercept):
 [-16.1260823]
array([1, 0, 1, 1, 0, 0, 0, 1, 0, 1, 0, 1, 0, 1, 1, 0])
```

```
#Evaluation metrics
from sklearn.metrics import accuracy_score, confusion_matrix, classification
accuracy = accuracy_score(y_test, y_pred)
print('Accuracy Score:', accuracy)
print('Confusion Matrix:\n', confusion_matrix(y_test, y_pred))
print('\nClassification Report:\n', classification_report(y_test, y_pred))
```

```
Accuracy Score: 0.6875
Confusion Matrix:
 [[4 1]
 [4 7]]

Classification Report:
              precision    recall  f1-score   support

           0       0.50      0.80      0.62         5
           1       0.88      0.64      0.74        11

    accuracy                           0.69        16
   macro avg       0.69      0.72      0.68        16
weighted avg       0.76      0.69      0.70        16
```

```
#Prediction
predict=classifier.predict([[22, 1, 8, 1, 1, 1, 0, 0, 1, 0, 0]])
print("Single Prediction (Class 0 or 1):", predict)
```

```
Single Prediction (Class 0 or 1): [1]
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: Use
  warnings.warn(
```

```
#deployment
import pickle
filename="placement_selection_model.sav"
pickle.dump(classifier,open(filename,'wb'))
```

26	22	Male	Electrical	1	8	0	0	0
27	22	Male	Information Technology	1	8	0	0	0
28	22	Male	Information Technology	1	8	0	0	0
32	21	Female	Civil	1	7	0	0	0
33	21	Male	Electrical	0	7	0	0	0
34	21	Male	Computer Science	1	6	0	0	0
35	21	Male	Information Technology	1	6	0	0	0
36	22	Male	Civil	0	7	0	0	0
37	22	Male	Computer Science	1	8	0	0	0
38	22	Male	Information Technology	1	6	0	0	0

```
#Use in another file
loaded_model=pickle.load(open("placement_selection_model.sav",'rb'))
predict=loaded_model.predict([[22, 1, 8, 1, 1, 1, 0, 0, 1, 0, 0]])
print("Loaded Model Prediction:", predict)
```

```
Loaded Model Prediction: [1]
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: Use
  warnings.warn(
```

```
# Get values from the user
age = int(input("Enter your Age: "))
internships = int(input("Enter number of Internships: "))
cgpa = float(input("Enter CGPA: "))
hostel = int(input("Enter Hostel (0 or 1): "))
history_of_backlogs = int(input("Enter History of Backlogs (0 or 1): "))
gender_male = int(input("Enter Gender (1 for Male, 0 for Female): "))
stream_cs = int(input("Enter Computer Science Stream (0 or 1): "))
stream_ee = int(input("Enter Electrical Stream (0 or 1): "))
stream_ece = int(input("Enter Electronics and Communication Stream (0 or 1):
stream_it = int(input("Enter Information Technology Stream (0 or 1): "))
stream_me = int(input("Enter Mechanical Stream (0 or 1): "))

# Pass user inputs to model prediction
predict = loaded_model.predict([[age, internships, cgpa, hostel, history_of_
print("Predicted Class for input user:", predict)
```

```
Enter your Age: 34
Enter number of Internships: 3
Enter CGPA: 89
Enter Hostel (0 or 1): 1
Enter History of Backlogs (0 or 1): 0
Enter Gender (1 for Male, 0 for Female): 0
Enter Computer Science Stream (0 or 1): 1
Enter Electrical Stream (0 or 1): 0
Enter Electronics and Communication Stream (0 or 1): 0
Enter Information Technology Stream (0 or 1): 0
Enter Mechanical Stream (0 or 1): 0
Predicted Class for input user: [1]
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: Use
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```

```
pip install streamlit
```

```
Collecting streamlit
  Downloading streamlit-1.66.0-py3-none-any.whl.metadata (10 kB)
Requirement already satisfied: altair!=5.4.0,!=5.4.1,<7,>=5.0.0 in /usr/local
Requirement already satisfied: click<9,>=7.0 in /usr/local/lib/python3.13/dis
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Requirement already satisfied: packaging>=20 in /usr/local/lib/python3.13/dis
Requirement already satisfied: pandas<4,>=1.5.3 in /usr/local/lib/python3.13/
Requirement already satisfied: pillow<13,>=9.2.0 in /usr/local/lib/python3.13
Collecting pydeck<1,>=0.8.0b4 (from streamlit)
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Requirement already satisfied: protobuf<8,>=5.26.1 in /usr/local/lib/python3.
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Requirement already satisfied: rpds-py>=0.25.0 in /usr/local/lib/python3.13/d
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.13/dist-pac
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Installing collected packages: pydeck, streamlit
Successfully installed pydeck-0.9.3 streamlit-1.66.0
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pip install numpy
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Requirement already satisfied: numpy in /usr/local/lib/python3.13/dist-packag
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!pip install streamlit
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Requirement already satisfied: streamlit in /usr/local/lib/python3.13/dist-pa
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```

```
pip install pyngrok
```

```
Collecting pyngrok
  Downloading pyngrok-8.1.2-py3-none-any.whl.metadata (8.6 kB)
Requirement already satisfied: PyYAML>=5.1 in /usr/local/lib/python3.13/dist-
Downloading pyngrok-8.1.2-py3-none-any.whl (25 kB)
Installing collected packages: pyngrok
Successfully installed pyngrok-8.1.2
```

```
%%writefile app.py
import streamlit as st
import pickle
import pandas as pd

# Set page configuration
st.set_page_config(page_title="Placement Selection Predictor", layout="cente

st.title("🎓 Student Placement Predictor")
st.write("Enter the student details below to predict placement status using

# Load the saved model
try:
    model = pickle.load(open("placement_selection_model.sav", "rb"))
except FileNotFoundError:
    st.error("Model file 'placement_selection_model.sav' not found. Please t
    st.stop()

# Create two columns for input fields
col1, col2 = st.columns(2)

with col1:
    age = st.number_input("Age", min_value=15, max_value=100, value=22, step
    internships = st.number_input("Number of Internships", min_value=0, max_
    cgpa = st.number_input("CGPA (out of 10)", min_value=0.0, max_value=10.0
    hostel = st.selectbox("Stays in Hostel?", options=[("No", 0), ("Yes", 1)
```

```
    history_of_backlogs = st.selectbox("Has History of Backlogs?", options=[

with col2:
    gender_male = st.selectbox("Gender", options=[("Female", 0), ("Male", 1)
    stream = st.selectbox("Academic Stream", options=[
        "Computer Science",
        "Electrical",
        "Electronics And Communication",
        "Information Technology",
        "Mechanical",
        "Other"
    ])

# Process the categorical stream into one-hot encoded variables matching tra
stream_cs = 1 if stream == "Computer Science" else 0
stream_ee = 1 if stream == "Electrical" else 0
stream_ece = 1 if stream == "Electronics And Communication" else 0
stream_it = 1 if stream == "Information Technology" else 0
stream_me = 1 if stream == "Mechanical" else 0

# Build the prediction dataframe with exactly matching feature names from tr
input_data = pd.DataFrame([[
    age, internships, cgpa, hostel, history_of_backlogs, gender_male,
    stream_cs, stream_ee, stream_ece, stream_it, stream_me
]], columns=[
    'Age', 'Internships', 'CGPA', 'Hostel', 'HistoryOfBacklogs', 'Gender_Mal
    'Stream_Computer Science', 'Stream_Electrical', 'Stream_Electronics And
    'Stream_Information Technology', 'Stream_Mechanical'
])

# Prediction trigger
if st.button("Predict Placement Status", type="primary"):
    prediction = model.predict(input_data)[0]

    st.markdown("--- ")
    if prediction == 1:
        st.success("🎉 **Predicted Status: Placed!** The student is likely
    else:
        st.warning("⚠️ **Predicted Status: Not Placed.** The student might
```

```
Overwriting app.py
```

```
!pip install streamlit pyngrok
```

```
Requirement already satisfied: streamlit in /usr/local/lib/python3.13/dist-pa
Requirement already satisfied: pyngrok in /usr/local/lib/python3.13/dist-pack
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